

The Adaptive Captain: A Live Win-Probability Engine for T20 Cricket

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Abstract

In modern T20 games captains often choose the next batter or bowler based on their intuition. The paper ([Ganesh, 2026](#)) models both the personnel decisions as a Markov Decision Process over match state, runs needed, balls remaining, wickets in hand, selecting whichever plan or order maximizes the simulated win or defend probability.

This project adapts that framework on real IPL data and tests the four extensions the paper ([Ganesh, 2026](#)) names as unbuilt. A sequence model and an adaptive batting policy produce real gains, contrastingly matchup conditioning and an adaptive bowling policy do not, and both failures are diagnosed. A venue-conditioned XI recommender is also built, with contested slots settled by simulation against a real opponent.

Repeating the paper's own headline test on the same real match gives a real gain, but far smaller than the paper's 4.1 percentage points, traced to this project training on eight IPL seasons against the paper's all seasons.

1 Introduction

In modern day T20 cricket, the decisions like who bats next after a wicket or who bowls the next over are still driven by intuitions than by data. Captains makes these decisions under acute time pressure, typically guided by gut feeling and mental model built from career statistics. Current cricket analytics mainly focuses on prediction and rarely extends to personalized recommendations ([Wickramasinghe, 2022](#)). Season-long aggregate and overall strike rate and economy rate often hide the information that matters the most for an in match decisions. For instance, Daniel Sams's overall economy is 8.20 which is perfectly solid but his death over economy is 13.18, nearly 5 runs gap. So what matters most is the perfect window to use his spell.

The match state depends on three important variables runs remaining / projected target, balls remaining, wickets in hand and this applies for both batting and bowling team. Based on these variables the batting team captain should make decision depending on how many batsmen left considering the aggression against current match state. Similarly, the bowling team captain must rotate the bowlers subject to their quota of 4 overs and consecutive

over rule, while accounting for phase specific economy and wicket taking ability. These decisions are therefore naturally modeled as Markov Decision Processes ([Bellman, 1957](#)) ([Puterman, 1994](#)), with the objective of maximizing a probability that runs scored to get the target, or that runs conceded to defend it. This project address the gap in previous works that uses Markov Decision Process ([Ganesh, 2026](#)) these three limitations; current simulations are performed based on static historical profiles, precommitted personnel decisions like next batter after a wicket and fixed bowler rotation plan, ball by ball independence where no momentum or streak is modeled. This work builds the same base Monte Carlo simulation, same estimator (Laplace smoothing + James-Stein shrinkage) and same search methods for making these decisions, then targets the limitations. It mitigates the ball by ball independence limitation by a trained GRU sequence model wired as a ratio adjustment on the player's own base rate ([Kyunghyun Cho, 2014](#)). Then the limitation precommitted personnel decisions is solved by re scoring all eligible candidates at a particular decision point for every trail by their expected phase performance adjusted for that trial's own current scoring pressure, and picks the best-scoring one on the spot rather than committing to one fixed order or plan in advance. Finally, addressing the limitation of static historical profile by blending recent season rate with the career rate of each player by using James Stein shrunk estimate weighted by how much recent data they actually have, so current form counts without discarding career sample size ([James and Stein, 1961](#)). Then this project builds a venue conditioned playing XI recommender which is a phase aware venue fit adjustment computed from each venue's own real, change point fitted phase structure, an experience factor, and a four-signal temperament score combine into one player value, and the final XI is build by simulating the candidates in a full match against the real opponent own historical XI.

2 Background

2.1 T20 primer

T20 cricket is a limited overs format with two innings, and each innings consists of 20 overs. A toss

decides who bats first, and the first batting team sets a target, and the chasing team tries to chase the target with in 20 overs. A over is a set of 6 legal balls, and 20 over consist of 120 balls. There should be at least 5 players who can bowl and each bowler gets the quota of 4 overs each. The term wicket is used when a batsman gets dismissed and the next batsman comes to bat. Other terms are run rate - average runs scored in a over. Strike rate - runs per 100 balls. Economy rate - average runs conceded per over.

2.2 Simulation-Based Optimization of Batting Order and Bowling Plans

This project addresses the limitations identified in (Ganesh, 2026). To enable a direct comparison with the baseline, this work adapts the unified MDP framework from (Ganesh, 2026), which represents both batting order and bowling plan with in a shared state space. The work also uses the concepts of Laplace smoothing for handling sparse outcomes (Laplace, 1814) and James-Stein shrinkage for stabilizing the player specific rates during the simulation process. The personnel decisions are found using exhaustive search over all orderings for batting, simulated annealing for bowling plans (Kirkpatrick and Vecchi, 1983).

2.3 State-Space Precedent in Limited-Overs Cricket

The three state dependent variables - remaining balls, runs required, wickets remaining are introduced by (Duckworth and Lewis, 1998), which is the standard method for resetting targets for limited over matches that are affected by rain. DLS uses these variables implicitly to build its resource table.

3 Algorithms

3.1 Pipeline Overview

The pipeline helps to understand the whole system at once, where every component is a stage in the pipeline rather than a standalone tool. The base pipeline has three stages. Building the player profile directly from the real training season ball by ball data is the first stage. The hard zeros in the player profile is removed using Laplace smoothing and James Stein shrinkage is used to blend each players own rate towards the population average if their sample is thin. The output is an outcome distribution per (player, phase) which is used as a lookup table in the up coming stages. That ta-

ble is then used by the simulator. Starting from a match state, the simulator draws each ball from the relevant player's profile until the innings ends naturally, that is either the target is reached, all wickets fall, or the overs run out. Repeating this process many time and averaging the outcomes is the Monte Carlo estimate of a decision value. There are two search algorithms that calls this simulator repeatedly for each candidate decision. The first search algorithm is the exhaustive search that tries every possible batting order and returns the best. The second search algorithm is simulated annealing that proposes and scores the candidate bowling plans. The result of the previous stage is used in the current stage.

Some plugins are attached to the base pipeline to improve the quality of the result. The first plugins are sequence model and momentum. They both get attached to the simulator adjusting the probability distribution the simulator draws from at each step. this is carried out without changing anything about the profiles or the search algorithms around them. The second plugin is adaptive policy, that changes who the simulator promotes next instead of following a fixed batting order or bowling policy. With this simulator picks the next batter or bowler at present. This is a change inside the simulator's own decision logic, not a new pipeline stage. Final plugin is recency blending that changes the player profile itself before it ever reaches the simulator. this blends a recent seasons window into the career profile using the same James Stein formula, so the simulator gets biased data towards recency.

Along with it there is a framework to make playing XI selection. The framework never calls the simulator instead it uses its own greedy algorithm and venue adjusted value formulas to finalize the playing XI. The real reason the framework did not use the simulator is due to computational limitation. Picking 11 from a squad of 25 results in millions of combinations and ideally solving XI selection, batting order and bowling plan jointly makes it a larger problem. Therefore the XI selection remains outside the pipeline.

There is one piece that sits in downstream rather than inside the simulator is the XGBoost win probability classifier. The simulator produces a raw final state like runs scored, wickets lost, balls used and the XGBoost model converts this state into a win probability number. It is used in exactly two places in this project: scoring the batting side's decisions and, much later, flagging which real historical balls

were genuinely uncertain for the temperament signal.

3.2 State Space and Match Dynamics

The match state at any ball is a triple $S = (r, b, w)$: r is the runs required to reach the target; b is the remaining legal balls left in this innings; w is the wickets in hand in batting perspective and wickets taken in bowling perspective. The Markov property is an assumption where the probability distribution over all future trajectories from this point depends only on S and not on the sequence of deliveries that produced it.

On each legal delivery exactly one of the seven outcomes $o \in \Omega = \{W, 0, 1, 2, 3, 4, 6\}$ occurs, here W denotes the wicket. The outcome of $o = 5$ is excluded as it is very rare occurrence under a particular conditions. The (Ganesh, 2026) own finding states that this occurred under 0.1% of the deliveries.

3.3 Player Profile Estimation

3.3.1 Outcome counting and wicket attribution

For each player or phase, raw outcome counts are accumulated over all legal deliveries involving that player. Here two distinct attribution rules apply, they are - a batsman is given out only the player got dismissed and this excludes the non striker run out from their count. Similarly, the bowler is credited with a wicket only if dismisses a batsman except a run out.

3.3.2 Laplace smoothing

A player who has never produced certain outcome, for instance a number 5 batsman never hit a six in the power play, would receive a 0 probability estimate for that outcome. This will result in biasing simulated win or defend probability. Laplace smoothing mitigates this problem by adding 1 to every outcome's raw count before turning them into probabilities, such that no outcome ever gets 0 probability estimate. (Laplace, 1814)

$$\tilde{c}_i^{o,\phi} = c_i^{o,\phi} + \alpha, \quad \alpha = 1, \quad \hat{p}_i^{o,\phi} = \frac{\tilde{c}_i^{o,\phi}}{\sum_{o'} \tilde{c}_i^{o',\phi}}$$

3.3.3 James Stein shrinkage

Many players have limited exposure to certain phase and role combination. The Laplace smoothed individual estimate for those players are high in

variance estimate from few observations. James Stein shrinkage addresses this by blending a noisy individual estimate with a group average, weighted by how much real data backs the individual estimate. (James and Stein, 1961)

$$\lambda_i^\phi = \frac{n_i^\phi}{n_i^\phi + n_{\min}}, \quad n_{\min} = 50,$$

$$\mathbf{p}_i^\phi = \lambda_i^\phi \hat{\mathbf{p}}_i^\phi + (1 - \lambda_i^\phi) \bar{\mathbf{p}}^\phi$$

A player with zero historical data collapses completely to population average (

$$\lambda = 0$$

), as their real sample grows their own estimate dominates.

3.3.4 Fixed phase scheme

This project uses a fixed three phase scheme - Power play which comprises of over 0 to 5, then middle overs ranging from 6 to 14, and death over from 15 to 19, which is identical at every venue.

3.4 Monte Carlo Simulation

For a fixed batting order or a bowling plan, the value at the initial state is estimated as $\hat{V} = \frac{1}{N} \sum_{n=1}^N 1[\text{trajectory}^{(n)} \text{ wins/defends}]$. This is an unbiased estimator which has binomial standard error that is bounded by $1/(2\sqrt{N})$.

To make the process fast, all simulations are run in parallel using vectorized operations. At each ball, the simulator samples outcomes for every active trial at the same time from precomputed probability tables. Simulations that have already ended are masked, so they do not receive any further updates.

3.5 Exhaustive Search — Batting Order

To select the batting order, the model considers up to 6 remaining candidates. With 6 players there are at most $6! = 720$ possible batting orders, extending this to at least one candidate result is 5040 combinations which is computationally heavy. The method uses two stages, stage 1 a quick screening. The model runs a small number of simulation say 3000 trials for every possible batting order and identifies the most promising ones. Then in stage 2 model runs many more simulation say 20000 trials only for the best 10 orders from the stage 1, giving a more accurate estimate of their win probabilities. This approach saves time because the system does not spend high computational effort evaluating clearly weaker batting orders.

3.6 Simulated Annealing — Bowling Plan

For making a bowling plan selection, the problem is much larger. A realistic death game scenario can have many valid bowling plans even after applying rules such as limited over bowler quota and over rotation rule. Testing every possible plan with multiple simulations would be computationally heavy.

As a remedy, this project uses simulated annealing (Kirkpatrick and Vecchi, 1983) an optimization algorithm that searches for better plans. Initially, it starts with one bowling plan, and then make small changes like replacing the bowler for one over with another one who is eligible to bowl. if this new plan has higher chance of winning estimation, then it is accepted, else it may still be accepted occasionally. The chance of accepting the plan is $\exp(\Delta V/T)$, where ΔV is the change in estimated win percentage and T is the temperature. The temperature starts at 0.05 and slowly decays as the steps increases which results in less acceptance of poor estimate plan near the end.

3.7 Sequence Model — GRU

This project uses a single layer Gated Recurrent Unit (Kyunghyun Cho, 2014) to mitigate the limitation of Ball by ball independence (Ganesh, 2026). the model takes last 6 ball's outcome and the current phase as features and trained with Adam optimizer, then cross entropy loss is computed over 113000 real deliveries. This is then used in the same way as a ratio adjustment against the population baseline for that phase, multiplied onto the player's own base rate.

3.8 Adaptive Decision Policy

Another limitation that is mentioned in (Ganesh, 2026) is precommitted personnel decisions. This project addresses it by building a tractable, one step approximation rather than the deciding fully. When a wicket falls the model chooses the next batter and at the start of each over the next bowler is selected. this decision is solely depends on the current situation in that individual simulation. For instance, if the batting side is under high scoring pressure, the model may prefer a batter who can score quickly, even if that player has a higher chance of getting out.

Shared pressure term (same for batting and bowling):

$$\text{pressure} = \text{clip} \left(\frac{\text{required run rate}}{\text{phase-average run rate}}, 0.5, 2.5 \right)$$

The pressure value is limited to between 0.5 and 2.5 such that rare match situations are not expected to produce extreme decision. For the batting decisions, the model balances the expected runs from a batter, then the risk of losing a wicket and the current scoring pressure. For bowling decisions, it uses the same idea but from the bowling team's perspective, that is, it prefers bowlers who can reduce runs and increase wicket-taking chances, especially when the batting side is under pressure.

Batting score, for each candidate batter:

$$\text{score} = E[\text{runs}] \times (1 + w \cdot (\text{pressure} - 1)) - P(\text{wicket}) \times \frac{2}{\text{pressure}}$$

Bowling score, for each candidate bowler:

$$\text{score} = -E[\text{runs conceded}] \times (1 + w \cdot (\text{pressure} - 1)) + P(\text{wicket}) \times (2 \times \text{pressure})$$

3.9 Recency-Blended Profile

The final limitation that is mentioned in the (Ganesh, 2026) is static historical profile for players. This is addressed by building a recent window profile and blending it with the unchanged career profile. For building this window last three seasons 2023, 2024, 2025 are taken into account. Once the window is build using the same James-Stein formula (James and Stein, 1961) that is used in player profile estimation is used again here to blend the recent window profile with the career profile.

3.10 XGBoost Win-Probability Model

This project uses XGBoost a pretrained classifier that predicts the $P(\text{win})$ from the match state features. The model is trained on real historical chase data from second innings. Holdout method is used training this model using season before 2026 and season 2026 is used for testing.

The XGBoost model is preferred over the logistic regression because gradient boosted trees can capture nonlinear interactions between chase state, venue, and player tendencies which a linear model can not. The XGBoost model is trained with multiple features that are grouped into three. First, baseline features like runs required, balls remaining, wickets remaining, required run rate, current run rate, required rate minus current rate are necessary features that serves as a base for the

model. Secondly, Venue based features like rolling par score for 2nd innings at this venue, historical chase success rate at this venue, Dew severity, average bounce, average turn. Finally, tendency based features the specific batter’s own phase strike rate, batters boundary dependency, batters strike rate against pace and spin, bowlers phase economy, bowlers dot ball percentage in a phase, the phase information.

The model functions by adding many shallow decision trees one after another, where each tree corrects the previous trees mistakes. Then the final output is passed through a sigmoid function to give a bounded probability between 0 and 1. Hyper parameters used in this model are 300 trees, maximum depth of 4 and learning rate 0.05.

3.11 Venue Conditioned XI selection

The paper (Ganesh, 2026) uses a fixed XI already on the field, but there is high chance that the XI might not be optimal to that venue. This project builds a venue conditioned Playing XI recommender by combining every players own recency blended batting or bowling value, followed by a real bounded venue fit adjustment, along with players experience factor and finally temperament signal and the XI highest-value players who satisfy the real IPL squad rules are picked as the recommended XI.

Each venue’s own dynamic phase boundaries are identified directly from that venue’s own over by over history. Which is a change point segmentation over six tactical features per over, run rate, wicket rate, pace versus spin mix, boundary rate, dot ball rate, and strike rotation rate, then chooses between three to five contiguous phases per venue using a BIC-style penalty which is a penalty on model complexity. It rewards a segmentation for fitting the data well, but penalizes for every extra phase it adds, so the algorithm does not just keep splitting into more and more phases to fit noise. Within each of a venue’s own phases, this project then works out that phase’s own real spin versus pace wicket ratio. This is because a venue’s preference for spin or pace can vary throughout the innings. Real numbers for two venues are follows.

Table 1: Phase-wise spin-wicket ratios at MA Chidambaram Stadium.

Phase	Overs	Spin-wicket ratio
P1	0–2	0.227
P2	3–5	0.373
P3	6–12	0.670
P4	13–16	0.521
P5	17–19	0.063

Table 2: Phase-wise spin-wicket ratios at M Chinnaswamy Stadium.

Phase	Overs	Spin-wicket ratio
P1	0–5	0.219
P2	6–11	0.600
P3	12–16	0.461
P4	17–19	0.084

The two venues split into a different number of phases at different over ranges, but still they both conclude with a spin heavy middle overs and an almost entirely pace driven death, which is exactly why one flat number for the whole innings cannot serve every venue.

A batter’s phase aware value at a venue is a weighted average across that venue’s own phases, where each phase is weighted by number of overs it spans. Within a single phase with spin ratio s , a batter’s value is

$$SR_{\text{blended}} = (1 - s) SR_{\text{pace}} + s SR_{\text{spin}}$$

Similarly a bowler’s phase aware value follows the same shape in reverse:

$$V_{b,p} = \frac{1}{E_{\text{con}_{b,p}}} (0.75 + 0.5 F_{b,p}),$$

The fit score is the phase’s own spin ratio when the bowler is a spinner

$$F_{b,p} = s_p \quad \text{if bowler } b \text{ is a spinner}$$

The fit score for phase’s own pace ratio when the bowler bowls pace is

$$F_{b,p} = 1 - s_p \quad \text{if bowler } b \text{ is a pace bowler}$$

Both formulas use the (James and Stein, 1961) shrinkage rule as every other estimate in this

project, blending a player’s own thin samples toward the population average rather than trusting a handful of balls outright.

Experience nudges this value up slightly. A player’s experience score is

$$E_i = \min \left(1, \frac{B_i}{Q_{0.75}(B)} \right),$$

so a handful of ultra veterans cannot dominate this factor once they clear that saturation point. This is gives at most a 15 percent boost

$$V_{i,p}^{\text{adj}} = V_{i,p} (1 + 0.15E_i),$$

Temperament nudges the value up or down again, from four independent pressure signals, each shrunk toward neutral when data is thin so a small sample cannot swing the score. Balance compares a player’s second-innings rate on genuinely uncertain balls, win probability between 0.35 and 0.65, against their own overall second-innings rate, blended toward 1.0 by a weight

$$\text{Balance}_i = w_i \left(\frac{R_{i,2\text{nd}}^{\text{uncertain}}}{R_{i,2\text{nd}}^{\text{overall}}} \right) + (1 - w_i),$$

$$w_i = \frac{n_i}{n_i + m}.$$

Carry compares a player’s own contribution against their teammates’ average in the same innings. Scarcity makes the same teammate comparison but only in matches where the run rate fell below that venue’s own historical median. Then Gear shift which is applicable only to batters, checks for a strike rate and acceleration later in the innings, or a fast start that beats the venue and phase average. Each of these three needs at least two qualifying instances to register at all, and then adds up to 15 percent to a player’s score in proportion to how often the player stood out. All the signals that apply to a player’s role are averaged into one combined temperament multiplier. Real numbers for recognizable players - table 3 and table 4

Dhoni’s Balance signal sits slightly below neutral at 0.949 because it reflects smart strike rotation rather than aggressive hitting under pressure, which captures players style.

Putting the three adjustments together, a player’s final value is

$$V_{i,p}^{\text{final}} = V_{i,p}^{\text{phase}} (1 + 0.15E_i) T_i,$$

The eleven players are chosen by a greedy rule which ranks every eligible player by this final value. A real recommended XI, Mumbai Indians at Wankhede Stadium, 11 players, 4 overseas, 2 keepers, 5 bowling-capable:

Table 5: Player roles and phase-aware values.

Player	Role	Country	Bat value	Bowl value	Total value
RD Rickelton	Wicketkeeper	South Africa	222.88	–	222.88
SA Yadav	Batter	India	196.49	–	196.49
Q de Kock	Batter	South Africa	195.41	–	195.41
RG Sharma	Top order Batter	India	195.32	–	195.32
Tilak Varma	Batting Allrounder	India	180.35	–	180.35
Naman Dhir	Top order Batter	India	175.84	–	175.84
HH Pandya	Allrounder	India	165.36	0.15	165.51
JJ Bumrah	Bowler	India	–	0.18	0.18
MJ Santner	Bowling Allrounder	New Zealand	–	0.15	0.15
TA Boult	Bowler	New Zealand	–	0.15	0.15
DL Chahar	Bowler	India	–	0.14	0.14

Bowling values are much smaller than batting value because it is built from economy rate rather than a strike rate, so the first ranked pass fills almost entirely with batters, then the bowlers are swapped ensuring at least five candidates bowlers make the XI

Tied to this is a validation step that sends the recommended XI through the full match simulator before treating it as final. The value formula only scores each player in isolation. It can not validate the performance of the XI against a specific real opponent. To validate this, the recommended XI is played out in a complete two innings match simulation, both sides batting and bowling against each other, at a real venue and against a real opponent’s own historical XI from the same season. Running this full simulation for every slot would be expensive, so only genuinely contested slots are checked this way a slot counts as contested when the value gap between the selected player and the next real alternative is small, and only for those slots are the two candidates actually simulated head to head against the same opponent to see which one gives the better real chance of winning. Slots where one player clearly out values the alternative are left as the value formula already decided them. This turns the value formula from the sole decision maker into a first pass, with the match simulator settling

Table 3: Temperament multipliers for selected batters.

Striker	Balance	Carry	Scarcity	Gearshift	Combined
MS Dhoni	0.949	1.086	1.083	1.132	1.063
RG Sharma	1.029	1.067	1.053	1.144	1.073
V Kohli	0.947	1.081	1.100	1.148	1.069

Table 4: Temperament multipliers for selected bowlers.

Bowler	Balance	Carry	Scarcity	Combined
DJ Bravo	1.211	1.065	1.058	1.112
JJ Bumrah	1.082	1.109	1.111	1.101

only the close calls, which is how the final recommended XI is actually formed.

4 Test

This project uses several testing approaches because the claims being evaluated are not all the same type, and no single method can fairly assess them all.

4.1 Data and Split

The data used in this project is real IPL ball by ball data based on Cricsheet and the venue based data is collected from web manually. This data is split using forward hold out method where every model and profile is built only from season before 2026 and evaluated on 2026 data. this mirrors the real deployment, where a captain only ever has past data available when making a decisions at present. Any feature that depends on recent form or previous seasons are built separately from each season using only data strictly before that season, so no seasons evaluation is ever contaminated by information from its own future.

4.2 Holdout log likelihood for outcome model claims

There are two extensions namely recency blending and the sequence model change the player outcome model itself. These extensions focuses on improving the probability of an event happening on the next ball. This is tested on 16,661 real held-out deliveries for each role. The average log probability of the model’s outcome that really happened is calculated, in which the value closer to zero means better calibration. This test checks the raw probability table against real match history, without adding Monte Carlo noise that could hide a real but small effect.

4.3 Simulated win or defend probability, for decision claims

This is a different family of extension where batting order search, bowling plan search and adaptive policy changes the decision but not the outcome. That is the changed decision improves the win probability. To validate this a real match state is simulated multiple times under each candidate decision and comparing the resulting probabilities. To evaluate this, several real match states are used. Match 1527674 represents an over-8 intervention, where the chasing side was 110/1 and needed 202 runs. Match 1527675 covers over 18, with the defending side need to defend 9 runs from 12 balls. Match 1527674 is a four-candidate batting-order case, built around Virat Kohli as the survivor, with 193 runs needed from 113 balls.

4.4 XI-selection validity

The playing XI selection is not evaluated by held out log likelihood or by simulated win probability. Precisely there is no ground truth available to pick a correct XI, since the recommendation itself a hypothetical selection and can not be verified. Instead the recommended XI is played out in a complete two innings match simulation, both sides batting and bowling against each other, at a real venue and against a real opponent’s own historical XI from the same season. Then identify the average winning probability with the recommended playing XI.

5 Results and Analysis

This section reports what each of those tests from previous section found. they are organized by evaluation method, so the result produced by same kind sits together.

5.1 Holdout log likelihood results

The two extensions, recency blend and the sequence model target the same underlying gap in the base model - a static, ball-independent outcome distribution that never adjusts for a player's recent form or the pattern of the last few balls. the sequence model shows some significant improvement while the recency blend still contributes a small. They both combine together to provide some additional gain. All three numbers following are scored on the same 16661 real held out deliveries per role. The combined recency + sequence model result

Table 6: Performance change from different modeling components.

Component	Batter Δ	Bowler Δ
Recency-blend alone	+0.0008	+0.0013
Sequence model alone	+0.0097	+0.0079
Combined	+0.0101	+0.0091

was checked directly rather than assumed. The sequence model's own evaluation logic already accepts any baseline profile bundle and blends its ratio adjustment on top of it, so passing it the recency blended bundle instead of the plain one produces the combined evaluation with no new logic required. Combining the two produces a small additional gain over the sequence model alone, raising the batter improvement from 0.0097 to 0.0101 and the bowler improvement from 0.0079 to 0.0091. This makes sense once the sequence model's own mechanism is considered: because it directly sees the last six real balls a player faced or bowled, it already captures most of what current form would otherwise convey, so a separate recency signal only has a small amount of genuinely new information left to contribute.

A model which predicted uniformly at random among the seven possible outcomes would score a log likelihood of -1.9459. But the baseline model scores -1.5156 for batters and -1.5188 for bowlers. This is a substantial improvement because it is achieved by applying Laplace smoothing and James Stein shrinkage techniques to real per phase outcome counts. Additionally, the entropy of the real population level outcome distribution can be computed directly for each phase, Firstly in the Power play the entropy is 1.4866 nats, then in the Middle overs it is 1.4646 and finally in the Death overs 1.6767. This number is the best average log

likelihood achievable using the population average for a particular phase, with no player-specific information. Moreover, it reflects a lower bound rather than a weakness in the estimator. It comes from the real, irreducible unpredictability of what happens on any individual cricket delivery. For an instance, a well played shot can still find a fielder, and a mistimed shot can reach the boundary. The baseline model already sits very close to this floor. Almost all of the information that is learnable about a cricket ball's outcome is already captured by understanding the phase alone, and a specific player's identity, their recent form, or the pattern of the last six balls are the left overs that adds a real but inherently narrow residual sitting on top of an already tight floor. This is the precise reason the extension deltas are all on the order of 0.01 rather than 0.1. They are a hard limit on how predictable any single delivery can be, with the base model already positioned close to that limit before any extension is even applied.

5.2 Simulated win or defend probability

These Extensions alters the decision instead of the outcome, where a particular instance from a match is simulated many times till the end to validate the decision made improved the win or defend probability more. For this testing two real matches anchor and three additional scenarios are built to test.

Table 7: Real and stylised cricket match scenarios.

Scenario	Type	Match	Situation
Kohli survives, RCB chase	Real	1527674, SRH vs RCB, M Chinnaswamy Stadium, 28 Mar 2026	193 needed off 113 balls
Short defend (2 overs)	Real	1527675, MI vs KKR, Wankhede Stadium, 29 Mar 2026	Defending 9 runs off the last 2 overs
High-pressure MI-style chase	Stylised	—	A tight, high-pressure chase
Comfortable CSK-style chase	Stylised	—	An already-winning chase
Longer defend (10 overs)	Stylised	—	A bigger defend, 6 bowlers

Picking the best batting order over a naive one resulted in +4.1 pp for (Ganesh, 2026) on KKR vs MI 29 March 2026 match. This project resulted in +0.59pp for the same match. This is mainly due to profile estimation difference, that depends on range of data used. This project trains on seasons 2018-2025 (492 matches), while the paper trains on 2008-2025 (1,161 matches) which is less than

half the data, so every player’s estimate is pulled harder toward the population average, compressing these gaps.

Choosing the next batter live beats the fixed order every time. But choosing the next bowler live does not on its own, but by adding the sequence modeling on top helps both batting and bowling.

Table 8: Scenario-level comparison of prediction approaches.

Scenario	Best fixed	Live choice	+ Sequence model
Kohli/RCB chase	0.0326	0.0464 (+0.0138)	0.1014 (+0.0550)
MI-style chase	0.0315	0.0369 (+0.0054)	0.0616 (+0.0244)
CSK-style chase	0.7575	0.7910 (+0.0335)	0.7749 (-0.0155)
Short defend	0.0540	0.0515 (-0.0025)	0.1037 (+0.0497)
Longer defend	0.1497	0.1474 (-0.0022)	0.2316 (+0.0820)

T20 matches often turn on a few percentage points, hence this gain is significant. A batting order or bowling plan can only rearrange the same 11 and it can not make any of them better. The batting order checks every possible order so the result is already the best possible one. the only improvement can be brought from choosing the candidates live. On the other hand the bowling plan search is a strong heuristic, but it lands close to the bound, which reasons the live bowling choice adds little. Moreover, a T20 innings contains of real, unavoidable ball-by-ball randomness, and 120 balls compounds it. At the trial counts used here (up to 50,000), the simulation’s own noise is about 0.22% , far smaller than every gain reported above, so these are real effects, not simulation noise.

6 Future work

For bowling a reinforcement learning methodology, replacing the current one step heuristic with a trained closed loop method capable of balancing immediate decisions against potentially better future outcomes. Additionally, the XI vs real opponent match simulation described previously have only run for the teams used as running examples. Therefore extending that same validation across all ten real IPL franchises, rather than a handful of them, remains open.

7 Conclusion

This work tests, not whether (Ganesh, 2026)’s method can be improved, but whether each of the four gaps the paper mentions can be built and whether building it helps. But that resulted a mixed conclusion. A sequence model and an adaptive batting policy substantially improve on the baseline contrastingly an adaptive bowling policy does not, on its own. Moreover matchup conditioning stays unsolved, blocked by real data sparsity. A venue-conditioned XI recommender, is also built with contested slots resolved by simulation against a real opponent’s historical XI. Repeating the paper’s own headline comparison on this project’s own data gives a real gain, but only around half a percentage point against the paper’s 4.1, traced to training on eight IPL seasons against the paper’s eighteen full season.

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